TFTN: A Transformer-Based Fusion Tracking Framework of Hyperspectral and RGB

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***Abstract*— Although the red, green, and blue (RGB) image has a high spatial resolution, it only depicts color intensities in RGB channels, which easily leads to the failure of the tracker based on RGB modality in some challenging scenarios, for example, when the color of the object and background is similar. The hyperspectral image with rich spectral information is more robust in these difficult situations, so it is essential to explore how to effectively apply hyperspectral features to supplement RGB information in object tracking. However, there is no fusion tracking algorithm based on hyperspectral and RGB data. Based on this, we propose a novel fusion tracking framework of hyperspectral and RGB in this article, termed as transformer-based fusion tracking network (TFTN), to enhance the performance of object tracking. Within the framework, we construct a dual-branch structure based on the Siamese network to obtain the modality-specific representations of different modality images. Besides, the framework is generic, which is suitable for the Siamese series of tracking algorithms. In addition, we design a Siamese 3-D convolutional neural network as the specific branch of hyperspectral modality for synchronous extraction of the spatial and spectral features of hyperspectral data, to give full play to the role of hyperspectral data in improving network tracking performance. Particularly, inspired by the structure of Transformer, we design a transformer-based fusion module to capture the potential interaction of intramodality and intermodality features of different modalities. This is the first work that combines the information of hyperspectral and RGB modalities to improve tracking performance. At the same time, it is also the first time that employs the self-attention module of Transformer to combine the information of different modalities for multimodality fusion tracking. Experimental results on the dataset composed of hyperspectral and RGB image sequences show that the proposed TFTN tracker is superior to the state-of-the-art trackers, demonstrating the effectiveness of this method.**

***Index Terms*— 3-D convolutional neural network (CNN), attention, fusion tracking, hyperspectral images (HSIs), transformer.**

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I. Introduction

OBJECT tracking is an important task in the field of remote sensing, which can be used on military and civilian sides, such as traffic management [1], route determination [2], and sea ice detection [3]. The purpose is to infer the tracked object's position in the subsequent frames as accurately as possible when the boundary frame of the interest object in the first video frame is given. Most tracking algorithms are studied for red, green, and blue (RGB) modalities. However, the RGB image has visual feature defects, which can easily lead to the drift of the RGB-based tracker in the case of background clutter (BC), occlusion (OCC), and so on. Therefore, supplementing other modality information to the RGB-based tracker will effectively improve the tracking performance.

With the advancement of spectral imaging technology, hyperspectral images (HSIs) with spatial and spectral information have gradually attracted attention. Compared with RGB images, when the appearance of different substances is similar, HSIs can provide more material information to improve the discrimination ability of different objects, as shown in Fig. 1. Although some works have preliminarily explored the application of HSIs in object tracking, the defects [such as low resolution (LR), etc.] caused by the imaging mechanism of HSIs easily lead to the failure of the HSI-based trackers. Therefore, it is necessary to combine the spatial and spectral information of HSIs and visual features of RGB images for object tracking.

However, how to effectively integrate the information of different modalities for tracking is challenging. At present, the feature-level deep learning-based fusion tracking algorithms have attracted much attention [4], [5], [6]. However, most of them use a simple concatenation method to combine multimodality features, ignoring the global deep interaction between different modality information. Transformer [7], [8] is a nonlocal model with self-attention and cross-attention layers, which can learn long-range dependencies and capture global receptive fields. It has shown good performance in some related works based on image-text [9], [10], [11], indicating that transformer has great application potential for multimodality and cross-modality tasks. Therefore, applying the transformer's structure to the fusion tracking field is worth exploring.

In addition, how to capture the specific information of hyperspectral modality is also very important for using hyperspectral data to improve the performance of the fusion tracking

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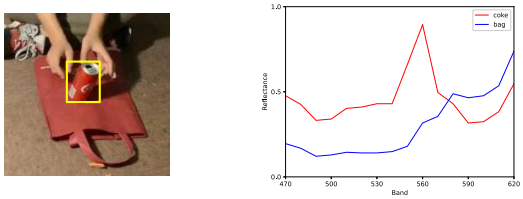


Fig. 1. Example of similar colors but distinct spectral information between the object and the background. On the left side, the relationship between a coke and a bag in the RGB image is displayed, where the coke is taken as the object, marked with a yellow box, and the bag is taken as the background. On the right side, the spectral responses at two pixels from the coke and the bag are displayed. Though two pixels are similar in color, their spectra are different.

task. Compared with RGB images, HSIs have a unique 3-D structure, which can have both spatial and rich spectral information. However, the research on making full use of hyperspectral information to improve object tracking performance is not enough. Therefore, to give full play to the role of hyperspectral data, it is necessary to explore a suitable network structure to obtain the modality-specific (MS) information of hyperspectral data with a 3-D structure. 3-D convolution kernel has natural advantages for extracting spatial and spectral information of hyperspectral data and is widely used in the tasks of HSI classification and detection. Therefore, how to effectively use a 3-D convolutional neural network (CNN) to obtain MS information of hyperspectral data for the fusion tracking task is worthy of focus.

To address the aforementioned issues, we propose a novel fusion tracking framework of hyperspectral and RGB based on Transformer, that is, transformer-based fusion tracking network (TFTN), which aims to improve the robustness of tracking by effectively using hyperspectral information to supplement the inherent defects of RGB modality. TFTN is a general framework suitable for the Siamese series of tracking algorithms. The framework contains two MS branches (one is the hyperspectral MS branch and the other is the RGB MS branch) which are used to process the two modality data, respectively, to obtain the MS representations of the two modalities. Each branch is designed based on the Siamese network. The hyperspectral MS branch uses a Siamese 3-D CNN to jointly learn 2-D spatial features and 1-D spectral features from hyperspectral data. These features obtained from the hyperspectral branch are unique to hyperspectral data, representing the specific representations of hyperspectral data in the task of multimodality fusion tracking. The RGB MS branch is mainly constructed by a Siamese 2-D CNN to fully obtain the information of RGB data. The rich visual features obtained by the RGB MS branch are viewed as the specific representations of the RGB modality. Then, a fusion module based on Transformer is designed to combine the MS representations of hyperspectral and RGB modalities. This module can adaptively capture the potential interaction between intramodality features and intermodality features of different modalities by the self-attention mechanism of the Transformer to predict the object's state effectively. The framework has an anchor-free structure, which reduces the dependence of the network on hyperparameters, making the tracking framework more concise. As far as we know, this is the first work that combines the information of hyperspectral and RGB modalities to improve tracking performance. Our major contributions are as follows.

1) We propose a novel multimodality fusion tracking framework, aiming to make the performance of the fusion tracker based on this framework higher than any single-modality tracker by fully using hyperspectral and RGB information. Particularly, the proposed framework is generic, which is suitable for the Siamese series of tracking algorithms.
2) We design a Siamese 3-D CNN as the hyperspectral MS branch, which uses the 3-D convolution kernel to slide jointly between the spatial and spectral dimensions of hyperspectral data, to fully obtain the specific representations of hyperspectral modality.
3) We also design a transformer-based fusion module (TFM) to explore the relationship between the MS representations of hyperspectral and RGB in this framework. To the best of our knowledge, it is the first time that employs the self-attention module of Transformer to combine the information of different modalities for multimodality fusion tracking.

The experimental results on the dataset composed of hyperspectral and RGB image sequences show that our method achieves leading performance, outperforming state-of-the-art trackers.

The remainder of this article is organized as follows. Section II briefly reviews some related works. Section III presents the proposed TFTN. Then, the experimental settings and the experimental results and analysis are presented in Sections IV and V, respectively. Finally, Section VI concludes this article.

II. Related Works

This section briefly reviews some related works to our research. This section introduces the object tracking work related to two modality data, respectively. Since there is no work that combines the information of hyperspectral and RGB modalities to improve the tracking performance, the fusion tracking work based on other multimodality data is introduced. In addition, our method uses the Transformer to combine different modality information, so we also briefly introduce the relevant content of the Transformer in this section. Therefore, this section includes four subsections: visual object tracking, hyperspectral object tracking, fusion tracking, and Transformer.

A. Visual Object Tracking

Visual object tracking plays a fundamental role in computer vision, which can be widely applied in the fields of intelligent monitoring [1], [12], automatic driving [13], robotics [14], and so on. However, it is challenging to perform robust object tracking due to some unpredictable factors, such as poor lighting conditions, OCC, and BC.

Many research methods [15], [16], [17] have been proposed to cope with these challenges. These methods can be mainly

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divided into two kinds: one is using correlation filters to obtain the features [18], [19], and the alternative, is using deep learning to extract features [20], [21], which has strong competitiveness.

The Siamese trackers [22], [23], [24] developed based on deep learning methods have aroused great attention for their superior performance recently. Bertinetto et al. [22] proposed the SiamFC tracker, which first extracted depth features by constructing Siamese networks for templates and search regions, and utilized a cross correlation layer (Xcorr) to transmit object information to search regions. After that, to enhance the tracking effect, Li et al. [25] proposed the SiamRPN tracker, which first introduced the regional recommendation network (RPN) of target detection into the tracking work and used the depth correlation to fuse the features for obtaining a precise object boundary box. Since then, many research works focus on how to improve Siamese trackers to obtain robust tracking results, such as developing the anchor-free structure [26], deepening the network architecture [27], and changing the way of information transmission [28]. At present, the trackers based on the Siamese structure are the most effective for visual object tracking, such as TransT [29], SiamCAR [30], SiamGAT [28], and SiamDW [31] trackers. However, the inherent defects of the visual image caused by the imaging mechanism still limit the further improvement of the performance of these trackers, such as the difficulty in distinguishing the object and the background when the object's color is similar to the surrounding's color in the visual image.

B. Hyperspectral Object Tracking

Compared with visual images, hyperspectral data has both spectral information and richer spatial information, which can provide more robust features for tracking tasks.

Hyperspectral object tracking is an emerging field. Some works [32], [33] have preliminarily explored the method of predicting the target position based on HSI videos. For example, in some early attempts, Nguyen et al. [34] ignored the structural characteristics of HSI, which only used spectral reflectance as the feature of object tracking. On the contrary, Uzkent et al. [35] proposed the DeepHKCF tracker which sacrificed valuable spectral information, converted HSIs into false-color images, and then input them into the deep convolution neural network for object tracking. However, these methods do not consider both spatial and spectral information, which undoubtedly limits the performance of hyperspectral object tracking. Different from the aforementioned methods, Xiong et al. [36] proposed the MHT method which used two visual feature descriptors, local spectral-spatial histogram of multidimensional gradients (SSHMG) and spatial distribution of materials, to characterize local spectral–spatial texture information and spectral composition abundance of HSIs, respectively. Then, two descriptors were further embedded into the background-aware correlation filter (BACF) to generate material-based tracking. In addition, Liu et al. [37] proposed the HA-Net method and Li et al. [38] proposed the BAE-Net method, which provided new ideas for the application of deep learning methods in the field of hyperspectral object tracking. Although the tracking algorithms based on HSIs have achieved preliminary progress, the inherent defects of hyperspectral modalities seriously limit the success rate of hyperspectral object tracking, for example, LR [39].

C. Fusion Tracking

Fusion tracking can boost the tracking robustness under adverse challenging conditions by leveraging the complementary information of different modality images. Therefore, we want to use hyperspectral data information to supplement the defects of RGB images to improve the tracking performance.

Most fusion tracking methods [40], [41] are studied for RGB and infrared image sequences to explore the combination methods of different modality features. Among these methods, the deep learning-based algorithms [42] are more competitive, because the powerful feature representation ability of deep neural networks can effectively learn the robust features of the object. There are three main kinds of fusion tracking algorithms based on deep learning: pixel-level fusion, feature-level fusion, and decision-level fusion. Xu et al. [43] and Zhang et al. [42] adopted the pixel-level fusion method to combine RGB and infrared modality information. The difference is that Xu et al. [43] simply used the infrared image as the fourth channel of the RGB image, while Zhang et al. [42] utilized the Siamese network to fuse multimodality images. Different from the pixel-level fusion method which combines different modality images in the input layer, the fusion method based on decision-level fuses the tracking results of different modalities in the part of the output. Lan et al. [40] used two trackers to track infrared and RGB images, and selected the results with larger confidence values of different modalities as the fusion results. In addition, the fusion tracking algorithm based on feature-level is more popular owing to its can fully explore the complementary information of multimodality at the feature layer. Zhang et al. [5] proposed SiamFT and DsiamFMT [6] methods are representative algorithms in feature-level-based fusion tracking, which achieved better tracking performance by adaptively fusing the convolution features of multimodality images. It is regretful that there is no fusion tracking algorithm based on pairs of hyperspectral and RGB image sequences, which promotes our exploration on how to effectively combine the modality information of hyperspectral and RGB for object tracking.

D. Transformer

Unlike most methods using CNN to fuse multimodality information, the Transformer model [7] can combine multimodality information by exploring the relationship between different modalities through the attention mechanism.

The Transformer model was first proposed for machine translation tasks in the field of natural language processing (NLP) and has been widely used in the field of computer vision recently [44], [45], [46]. Due to the Transformer model can focus on the global long-range information through the attention mechanism, the model has achieved good performance in many cross-modality and multimodality tasks, such

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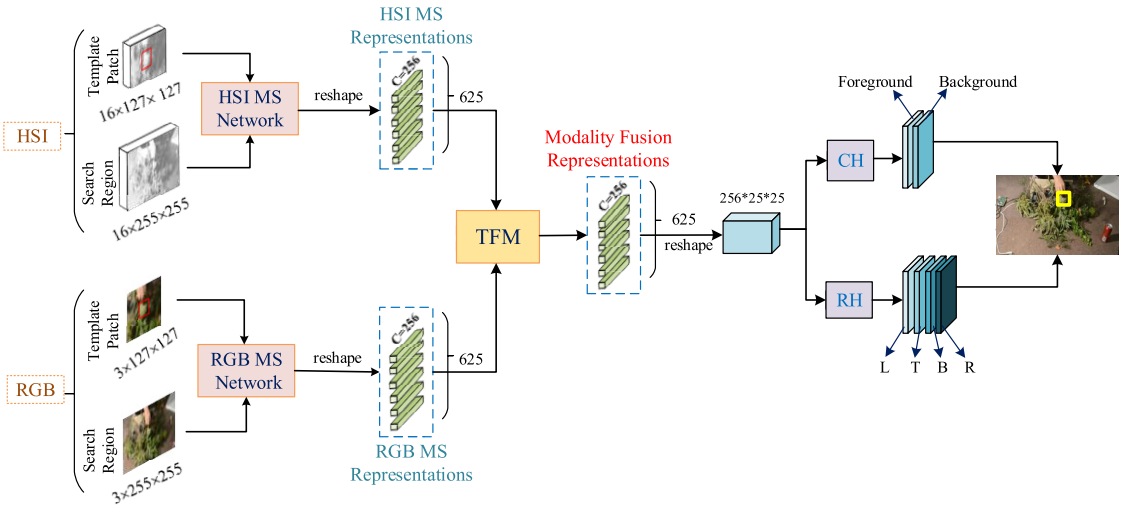


Fig. 2. Network architecture of the proposed TFTN. Classification prediction head (CH) represents the CH and regression prediction head (RH) represents the RH.

as the conversion of image subtitles [47], video retrieval [48], and RGB-D saliency detection [49]. However, there is no work on multimodality fusion tracking adopting the Transformer structure as far as we know. Therefore, we hope to apply the Transformer model to hyperspectral and RGB object tracking, which uses the attention mechanism in Transformer to capture the potential relationship between different modality information to improve the performance of object tracking.

III. Proposed Methods

The novel framework is proposed to achieve tracking by fusing the information of hyperspectral and RGB modalities in this section. First, the network architecture is introduced. Then, the specific content of the MS network is given. Finally, the TFM is described.

A. Network Architecture

The network architecture of the proposed TFTN is included in Fig. 2. We adopt an anchor-free architecture in this work that directly divides the tracking task into two subproblems as classification and regression to reduce the impact of hyperparameters on the tracking performance.

Two branches are constructed in the architecture to process hyperspectral and RGB modalities, respectively. Before entering the network branch, the multimodality images need to be preprocessed. In this work, the first frame in the video which contains the object state is chosen as the template image, and the rest frames of the video are the search images. The template patch which contains the information about the object and its local surrounding scenes is extended by twice the side length from the center of the object box in the template image and is reshaped to 127×127. The search region of each search image is expanded four times the side length from the center of the object in the previous frame and is reshaped to 255×255, covering the possible range of the object in the current frame typically. Then, the template patch and the search region are fed to the MS network to produce corresponding MS representations.

The MS representations of hyperspectral and RGB are obtained by two MS networks, respectively. We input two MS representations into the designed TFM to generate the effective modality fusion representations. Then the fusion representations are input into the classification head and the regression head, in turn, to solve the classification and regression problems of the tracking task. Eventually, the classification response map obtained by the classification head and the regression response map obtained by the regression head are used to jointly predict the state of the object in the current frame.

B. MS Network

The MS network is the fundamental structure of obtaining MS representations from different modalities based on the Siamese network. The MS network mainly consists of two simple subparts: one backbone part for feature extraction and the other part for transmitting the information of the template patch and the searching region.

1) Hyperspectral MS Network: Compared with the RGB image, the HSI with a 3-D structure has both spatial and richer spectral information. Therefore, hyperspectral data can provide more discriminative information for the network when the object and background have a similar appearance. Based on this, we design a Siamese 3-D CNN as the HSI MS network to fully obtain the HSI MS representations. The structure of the HSI MS network is shown in Fig. 3.

To better use hyperspectral data to improve the performance of the fusion tracking task, we design a multiscale 3-D CNN as the backbone of the HSI MS network to fully capture the multiscale spatial and spectral features of HSIs. The multiscale features refer to the information with different scale receptive fields, rather than the features with different sizes. The joint sliding of the 3-D convolution kernel between the spatial dimension and the spectral dimension of the HSI can extract more complex spatial–spectral joint information, improving the resolution of the network. In addition, the multiscale structure contains rich context information, which can better

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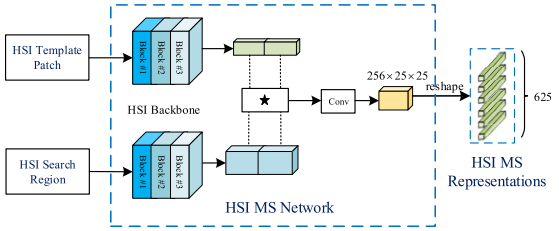


Fig. 3. Structure of the HSI MS network. The symbol ⋆ represents the cross correlation operation. There are two branches used to process the HSI template patch and the HSI search region, respectively. Then combine the information of the template patch and the searching region through ⋆ to obtain the HSI MS representations.

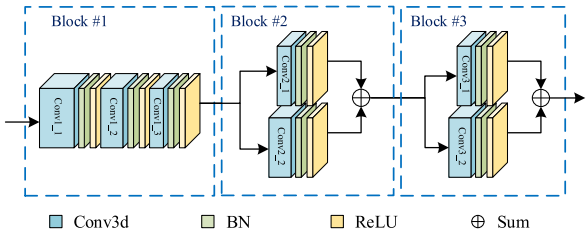


Fig. 4. Structure of the HSI backbone. There are three blocks in the network.

extract the features of spectral multiscale objects in the spatial domain. The multiscale 3-D CNN consists of three blocks, as shown in Fig. 4. Each block contains convolution layers, batch normalization layers, and activation layers. In addition, Block #2 and Block #3 contain summation operators, which can be used to capture spectral multiscale features better. Each block uses 3-D convolution layers that can extract spatial–spectral features of hyperspectral data naturally and elegantly. The detailed hyperparameter setting of this network is presented in Table I. The proposed multiscale 3-D CNN contains three blocks that can obtain more discriminative semantic information.

First, the HSI template and the HSI search region are input into two HSI backbones with the same structure to generate two feature maps $\varphi(z)$ and $\varphi(x)$. Among them, two HSI backbones share parameters. Then, to transmit the information from the template patch to the search region, $\varphi(z)$ is used as the kernel of $\varphi(x)$ to calculate the cross correlation of them. The cross correlation form is as follows:

$$R = \varphi(x) \star \varphi(z) \tag{1}$$

where $\star$ represents the cross correlation operation, and R represents the generated response map.

However, a cross correlation calculation can only generate a single-level-feature response map which is insufficient to fully represent modality information. Since different levels of features have different semantic information, we use the depth-wise cross correlation layer composed of two single-level cross correlation operations to calculate the correlation of the features extracted from the Block #2 and Block #3 of the HSI backbone. Then, the generated multiple semantic similarity maps with the same number of channels are spliced according to the channel dimension

$$R_{\text{all}} = \text{concat}(R_2, R_3) \tag{2}$$

TABLE I
PARAMETERS OF CONVOLUTIONAL LAYERS

Block Name	Kernel Name	Kernel Number	Kernel Size (H,W,B)	Kernel Stride Δ(H,W,B)
Block #1	conv1	256	7,7,5	2,2,3
	conv1_1	256	3,3,3	2,2,2
	conv1_2	256	3,3,1	2,2,1
Block #2	conv2_1	256	1,1,1	1,1,1
	conv2_2	256	1,1,3	
Block #3	conv3_1	256	1,1,1	1,1,1
	conv3_2	256	1,1,3	

The symbol B denotes the kernel size in the spectral dimension.

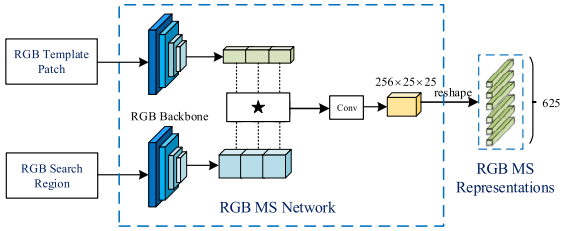


Fig. 5. Structure of the RGB MS network. The symbol ⋆ represents the cross correlation operation. There are two branches used to process the RGB template patch and the RGB search region, respectively. Then combine the information of the template patch and the searching region through ⋆ to obtain the RGB MS representations.

where R_2 and R_3 represent the semantic similarity maps of the Block #2 and Block #3, respectively. R_{all} represents the spliced response map.

In addition, in order to reduce the number of parameters, the 1×1 convolution kernel is used to reduce the dimension of the spliced semantic similarity map, and the reshaped dimension-reduction response map is adopted as the final HSI MS representations.

2) RGB MS Network: Similar to the structure of the HSI MS network, the RGB MS network also includes the parts of feature extraction and information transmission. RGB images have high spatial resolution and contain rich visual features (such as color, texture, etc.). Therefore, to obtain more effective RGB MS representations, we use a Siamese 2-D CNN to process the RGB data. The structure of the RGB MS network is shown in Fig. 5. In this work, we use the modified ResNet-50 as the same in [27] as the RGB backbone. In addition, consistent with the transmit information method of the HSI MS network, we also transform the information from the feature map of the template patch and that of the search region by calculating the depth correlation between them. We use three single-level cross correlation operations to calculate the correlation of the features extracted from the last three residual blocks of the backbone and splice the generated multiple semantic similarity maps according to the channel dimension

$$R_{\text{all}_{\text{rgb}}} = \text{concat}\left(R_{3_{\text{rgb}}}, R_{4_{\text{rgb}}}, R_{5_{\text{rgb}}}\right) \tag{3}$$

where $R_{3_{\text{rgb}}}$, $R_{4_{\text{rgb}}}$, and $R_{5_{\text{rgb}}}$ represent the semantic similarity maps of the third, fourth, and fifth residual blocks, respectively. $R_{\text{all}_{\text{rgb}}}$ represents the spliced response map. Finally, we reshape

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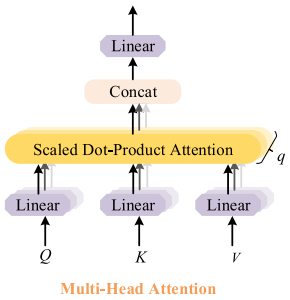


Fig. 6. Structure of multihead attention. It consists of several attention layers running in parallel. Each layer takes scaled dot-product attention.

the dimension of the spliced map and the reshaped map is adopted as the final RGB MS representation.

C. Transformer-Based Fusion Module

The key problem to performing tracking tasks using multimodality images is how to effectively combine different modality information. To solve this problem, we design a fusion module based on Transformer (TFM), which fully explores the intramodality and intermodality relationships of hyperspectral and RGB modalities through the attention mechanism of Transformer.

1) Attention of Transformer: The attention mechanism is an important component of the Transformer. The attention function is mapping query (Q), keys (K), and values (V) to an output. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. The scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{\text{QK}^T}{\sqrt{d_k}}\right)V \tag{4}$$

where d_k represents the dimension of input.

The Transformer model uses multihead attention to describe the global dependency between input and output. The structure of multihead attention is shown in Fig. 6. Multihead attention is a global receptive field in the region actually, which can focus on the information of different subspaces at different locations through multiple heads. All attention distributions are calculated by scale dot-product attention. The multihead attention is assumed to contain q heads, which is defined as

$$\text{MultiHead}(Q, K, V) = \text{concat}\left(H_1, \ldots, H_q\right)W^O, \tag{5}$$

$$H_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{6}$$

where $W^O \in \mathbb{R}^{\text{qd}_v \times d_m}$represents parameter matrices, $W_i^Q \in \mathbb{R}^{d_m \times d_k}$, $W_i^K \in \mathbb{R}^{d_m \times d_k}$, and $W_i^V \in \mathbb{R}^{d_m \times d_v}$ are the radiation projections of the parameter matrix of the ith head of Q, K and V, respectively.

2) Details of TFM: Inspired by the structure of Transformer, we can fuse multimodality features by calculating the cross-attention from different MS representations, or calculating the self-attention of cascaded MS representations to achieve the purpose of multimodality feature fusion.

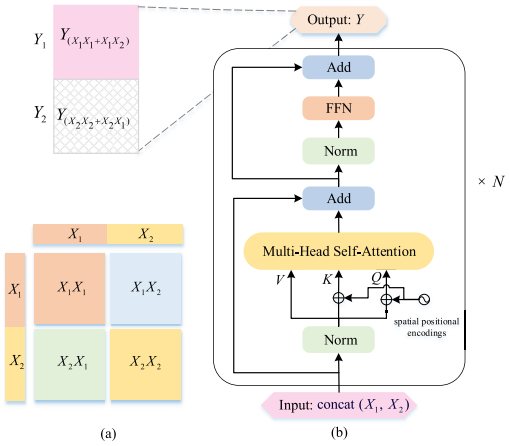


Fig. 7. Main idea of TFM is in (a) and the structure of the TFM is in (b), where X_1 is the RGB MS representations, and X_2 is the HSI MS representations.

However, compared with the multimodality fusion method based on cross-attention, the modality fusion method of the concatenation-based self-attention does not need to carefully design the fusion structure and automatically perform the information fusion within and between modalities, which greatly reduces the computational complexity. Therefore, TFM adopts the idea of concatenation-based self-attention fusion as the main idea of multimodality fusion, as shown in Fig. 7(a).

The structure of TFM is illustrated in Fig. 7(b). First, the RGB MS representations $X_1 \in \mathbb{R}^{\text{HW}\times C}$ and the HSI MS representations $X_2 \in \mathbb{R}^{\text{HW}\times C}$ are concatenated to get the input $X \in \mathbb{R}^{2\text{HW}\times C}$ of the fusion module

$$X = \text{concat}(X_1, X_2). \tag{7}$$

Then, we use the mechanism of multihead self-attention in the form of residual to adaptively integrate different aspects of the input information

$$\tilde{X} = X + \text{MultiHead}(X + P_x, X + P_x, X) \tag{8}$$

where $P_x \in \mathbb{R}^{2HW\times C}$ is the spatial positional encodings of X.

In addition, an FFN module which is a fully connected feed-forward network that consists of a linear transformation and a gaussian error linear unit (GeLU) is used to enhance the fitting ability of the model

$$\text{FFN}\left(\tilde{X}\right) = \text{GeLU}\left(\tilde{X}\right)W + b \tag{9}$$

where W stands for a weight matric and b represents basis vectors.

Finally, the output Y can be summarized as

$$Y = \tilde{X} + \text{FFN}\left(\tilde{X}\right). \tag{10}$$

In this work, we employ $H = W = 25$, $C = 256$, $q = 8$, and the number of TFM blocks $N = 8$.

The self-attention mechanism of Transformer focuses on the internal relationships of self-to-self. Based on this, the specific representations of different modalities are concatenated as the input of TFM, which can explore the relationship of intramodality and intermodality adaptively through the self-attention mechanism. Therefore, the final output result Y corresponding to the concatenated input can naturally be divided

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into two parts: one is the result Y_1 of the internal effect of RGB modality and the influence of hyperspectral modality on RGB; the other is the result Y_2 of the internal effect of hyperspectral modality and the influence of RGB modality on hyperspectral. Since we hope to use hyperspectral features to supplement RGB information, Y_1 is chosen as the final modality fusion representations.

IV. Experimental Settings

To test the performance of the proposed TFTN, experiments are conducted on the dataset composed of hyperspectral and RGB. In this section, the experimental details are introduced, including the implementation, dataset, and evaluation metrics.

A. Implementation Details

In this study, all experiments are conducted using a desktop equipped with an NVIDIA RTX 3090 graphics processing unit (GPU) and Intel Xeon Silver 4210R central processing unit (CPU). We adopt the modified ResNet-50 [27] as the RGB backbone to extract the deep features of the template and search region of the RGB modality. The backbone is initialized with the weights that are pretrained on ImageNet [50]. The initialization process of the hyperspectral branch is the same as that of the RGB branch, and the weight decay and momentum of the two branches are set to 0.0001 and 0.9, respectively. During the training process, the batch size is set as 16, and a total of 20 epochs are trained with stochastic gradient descent (SGD). For the first 5 warmup epochs, the learning rate linearly increased from 0.005 to 0.01 and then exponentially decayed to 0.0005 over a total of the rest 15 epochs. The processing speed of the program is 17.2 fps.

B. Dataset

The recently released dataset which includes various aligned hyperspectral and RGB video pairs is employed for training and testing in this study. As far as we know, this is the first public hyperspectral dataset published in the Hyperspectral Object Tracking Competition. The dataset includes hyperspectral videos, RGB videos, and false-color videos converted from HSIs. The HSI sequences which contain 16 bands in the wavelength from 470 to 620 nm are captured by the snapshot mosaic hyperspectral camera at video rate, and RGB videos are acquired in a close viewpoint as the hyperspectral videos. The dataset covers 11 challenging factors, including OCC, illumination variation (IV), BCs, scale variation (SV), deformation (DEF), motion blur (MB), fast motion (FM), in-plane rotation (IPR), out-of-plane rotation (OPR), out-of-view (OV), and LR. At present, some preliminary studies have been carried out on the dataset in the field of remote sensing. We use 40 video pairs composed of hyperspectral and RGB of the training set to train the network model and employ the testing set which includes 35 video pairs to evaluate the performance of the proposed method. Table II lists the sequence names and corresponding attributes of the testing set. In addition, we count the number of videos corresponding to different attributes in the testing set, as shown in Table III.

TABLE II
Sequence Names and Attributes of the Testing Set

Name	Attribute	Name	Attribute
Ball	SV, MB, OCC	Forest2	BC, OCC
Basketball	FM, MB, OCC, LR	Fruit	BC, OCC
Board	IPR, OPR, BC	Hand	BC, SV, DEF, OPR
Book	IPR, DEF, OPR	Kangaroo	BC, SV, DEF, OPR, MB
Bus	LR, BC, FM	Paper	IPR, DEF, OPR, SV
Bus2	IV, SV, OCC, FM	Pedestrian	IV, SV
Campus	IV, SV, OCC	Pedestrian2	LR, OCC, IV, DEF
Car	SV, IPR, OPR	Player	IPR, DEF, OPR, SV
Car2	SV, IPR, OPR	Playground	SV, OCC
Car3	SV, LR, OCC, IV	Rider1	LR, OCC, IV, SV
Card	IPR, BC, OCC	Rider2	LR, OCC, IV, SV
Coin	BC	Rubik	DEF, IPR, OPR
Coke	BC, IPR, OPR, FM, SV	Student	IV, SV
Drive	BC, IPR, OPR, SV	Toy1	BC, OCC
Excavator	IPR, OPR, SV, OCC, DEF	Toy2	BC, OCC, SV, IPR, OV,OPR
Face	SV, MB, IPR, OPR	Truck	OCC, OV, SV
Face2	IPR, OPR, SV, OCC	Worker	SV, LR, BC
Forest	BC, OCC		

TABLE III
Attribute Information of the Testing Set

Attribute	Video numbers	Attribute	Video numbers	Attribute	Video numbers
OCC	18	FM	4	IV	8
BC	14	OPR	15	IPR	14
SV	23	OV	2	DEF	8
MB	4	LR	7		

C. Evaluation Metrics

In this work, the area under the curve (AUC) score of the success rate plot and the precision rate at the threshold of 20 pixels (DP_20) value are utilized to evaluate the performance of the fusion tracking task.

When the overlapping rate between the bounding box generated by the tracker and the groundtruth is greater than a certain threshold, it means success. The overlapping rate (OR) is defined as

$$\mathrm{OR} = \frac{|A_1 \bigcap A_2|}{|A_1 \bigcup A_2|} \tag{11}$$

where A_1 indicates the groundtruth box and A_2 is the bounding box generated by the tracker. The symbol of $|\cdot|$ represents the number of pixels in the area. Success rate represents the percentage of successful frames to total frames. The success rate plot also shows the changing trend of success rate when the threshold changes from 0 to 1 at an interval of 0.02.

When the center position error between the bounding box predicted by the tracker and the groundtruth box is less than a certain threshold, it means precision. The center position error is represented by the average Euclidean distance, which is as follows:

$$D = \sqrt{(b_2 - b_1)^2 + (a_2 - a_1)^2} \tag{12}$$

where (a_1, b_1) is the center position of prediction and (a_2, b_2) is the center position of groundtruth. The precision rate represents the percentage of precision frames to total frames. The precision plot shows the trend of precision rate when the threshold changes from a small value to a large value.

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TABLE IV
PERFORMANCE OF 35 SEQUENCES. THE BEST THREE RESULTS ARE LABELED IN RGB, RESPECTIVELY

	HA-Net	BAE-Net	MHT	TransT	SiamGAT	SiamDW	SiamFC	ECO	fDDST	TFTN
AUC	0.693	0.614	0.592	0.687	0.636	0.547	0.521	0.570	0.467	0.764
DP_20	0.947	0.876	0.882	0.920	0.864	0.854	0.797	0.840	0.725	0.962
FPS	14.0	0.9	1.5	39.8	21.0	68.6	83.2	17.3	92.2	17.2

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experiment1: Performance Comparison With Advanced Trackers

Since this is the first fusion tracking based on hyperspectral and RGB data to test the performance of the TFTN tracker, we compare the TFTN tracker with nine state-of-the-art single-modality trackers on the testing set, including HA-Net, BAE-Net, MHT, TransT, SiamGAT, SiamDW, SiamFC, efficient convolution operators for tracking (ECO), and fDDST. In these trackers, HA-Net, BAE-Net, and MHT are designed for HSI tracking, while the others are only developed for tracking based on RGB images. Among them, the HA-Net tracker was developed for hyperspectral data in this dataset and won the Hyperspectral Object Tracking Competition. Besides, the TransT tracker is trained for massive optical datasets, which is representative of high-performance RGB trackers in recent years. The performance of TFTN is analyzed from both quantitative and qualitative aspects.

1) Quantitative Analyze: The quantitative results of ten trackers on each sequence of the testing set in terms of AUC and DP_20 are shown in Table IV. It can be seen that the proposed TFTN is superior to all compared trackers in successful performance and precision performance. Moreover, it is also clear to observe from Fig. 8(a) and (b) that the overall performance of TFTN outstrips all the compared trackers in terms of success rate and precision rate. All of the above indicate the effectiveness of TFTN in the fusion tracking of hyperspectral and RGB.

a) Comparison with RGB trackers: We compare the performance of the proposed TFTN with that of RGB trackers. From Table IV, we can observe that the overall performance of TFTN is 7.7% and 4.2% higher than that of TransT in terms of AUC score and DP_20 value, although TransT shows the best performance of RGB trackers in the above two aspects, respectively. It can be discovered from Fig. 8 that the TFTN achieves the best performance compared to that of the RGB tracker in most challenging scenarios, such as OCC, LR, IV, SV, DEF, and BC. Particularly, in the scenarios of IV, the performance of TFTN is 19.0% higher than that of the best comparative RGB tracker in terms of AUC score. It is exhibited that the proposed TFTN enhances the ability to cope with challenging scenarios by adding HSI information to the RGB modality can effectively improve the tracking performance. This is because HSIs can provide more robust features for RGB images under these challenges.

b) Comparison with hyperspectral trackers: It can also be seen from Table IV that the HA-Net performs best in the tracking based on HSIs, the overall performance of TFTN is 7.1% and 1.5% more than those of HA-Net in terms of AUC score and DP_20 value, respectively. In addition, Fig. 8 shows that TFTN outperforms all the comparative hyperspectral trackers in 11 challenging scenarios in terms of AUC score. The results show that the proposed TFTN makes up for the defect of the LR of HSI using the vision features provided by RGB images, enhancing tracking robustness.

2) Qualitative Analyze: The qualitative frame-by-frame comparison of trackers in terms of overlapping rate in some sequences is shown in Fig. 9. It can be seen that the proposed TFTN achieves a generally higher overlapping rate in these sequences compared with other trackers. In particular, the object is severely occluded in frames from 750 to 850 in Fig. 9(a), 250 to 350 in Fig. 9(b), and 310 to 360 in Fig. 9(e), respectively, while TFTN can still track it well. In addition, the proposed tracker also can run stably in these sequences on the whole.

Fig. 10 shows the qualitative tracking results of different trackers in some image sequences, which can intuitively compare the tracking performance of TFTN and other trackers. In these sequences, there are mainly the challenging scenes of OCC, BC, IV, LR, SV, DEF, and IPR.

Specifically, objects of four video sequences are faced with the challenge of obstacle OCC in the process of movement, and the resolution of the moving objects in Pedestrian2 and Car3 is also low. In addition, the lighting conditions of the object surrounding environment in Pedestrian2 (*#36*, *#279*, and *#339*) and Car3 (*#23*, *#73*, and *#158*) are poor, and the appearance and the color of the object and background in Card (*#238*, *#798*, and *#930*) and Toy1 (*#141*, *#210*, and *#253*) are similar. In particular, the Pedestrian2 sequences have the problem of object DEF, the Card sequences have the challenge of OPR, and the object scale in the Car3 sequences is also changing. Due to the interference of these challenging factors, most trackers are prone to tracking failure. In contrast, TFTN can still track the object robustly, indicating that the proposed method can effectively reduce the influence of the above challenging factors on the tracking results.

It can be clearly seen from the above examples that the proposed TFTN can effectively deal with various challenging scenarios, such as BC, LR, OCC, and IV, which fully demonstrates the competitive advantage of using hyperspectral features to supplement RGB information during tracking.

B. Experiment2: Effectiveness Evaluation of the Fusion Framework

The proposed fusion tracking framework, TFTN, includes a dual-branch structure based on the Siamese network and a

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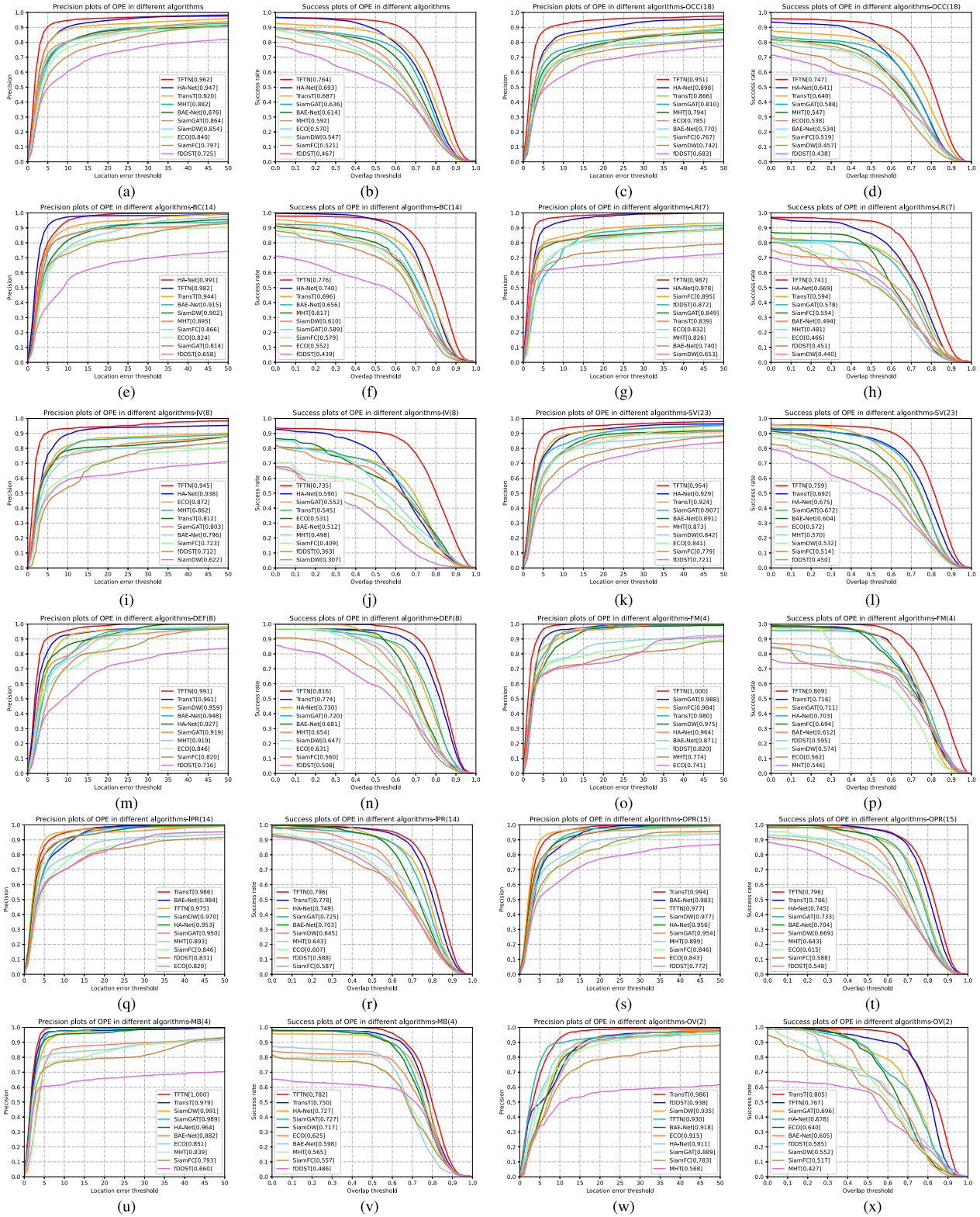


Fig. 8. Comparison of tracking results in terms of success rate and precision rate. (a) Precision plot of MB. (b) Success plot of MB. (c) Precision plot of OV. (d) Success plot of OV. (e) Precision plot of background clutters. (f) Success plot of background clutters. (g) Precision plot of low resolution. (h) Success plot of low resolution. (i) Precision plot of illumination variation. (j) Success plot of illumination variation. (k) Precision plot of scale variation. (l) Success plot of scale variation. (m) Precision plot of deformation. (n) Success plot of deformation. (o) Precision plot of fast motion. (p) Success plot of fast motion. (q) Precision plot of in-plane rotation. (r) Success plot of in-plane rotation. (s) Precision plot of out-of-plane rotation. (t) Success plot of out-of-plane rotation. (u) Precision plot of motion blur. (v) Success plot of motion blur. (w) Precision plot of out-of-view. (x) Success plot of out-of-view.

TFM, which are used to obtain different modality information and combine them to generate more robust features, so as to improve the performance of tracking tasks. To evaluate the effectiveness of the fusion framework, we test the performance of the model which the input of the dual-branch structure of the TFTN model is RGB data (TFTN_Dual_RGB)

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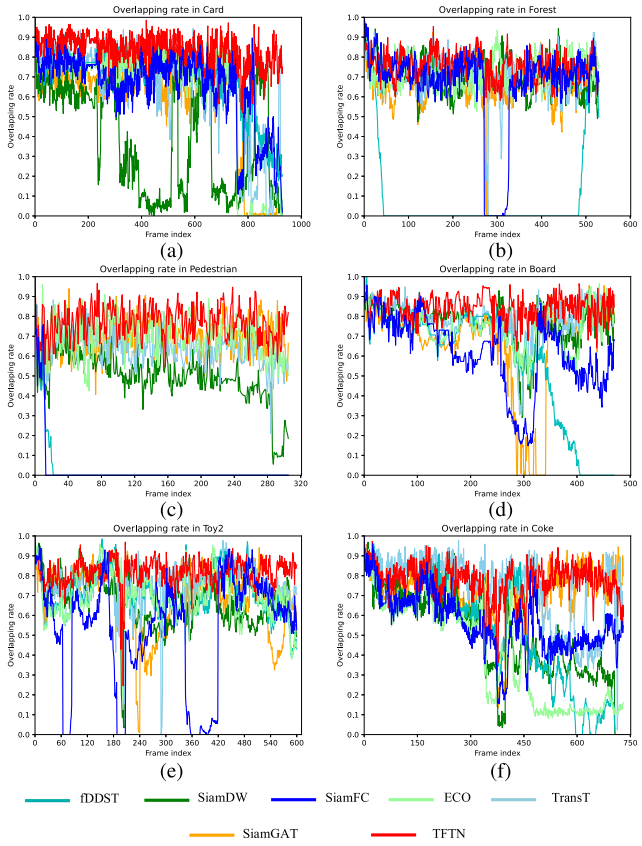


Fig. 9. Frame-by-frame comparison of seven trackers on six challenging image sequences in terms of overlapping rate. The horizontal axis shows the frame index and the vertical index indicates the overlapping rate. These six sequences from (a) to (f) are Card, Forest, Pedestrian, Board, Toy2, and Coke.

and test the performance of the model which only contains the RGB branch in the TFTN model (TFTN_single_RGB). In addition, we also test the performance of the TFTN model when the input of the dual-branch structure is HSI data (TFTN_Dual_HSI), and test the performance of the model which only contains the RGB branch in the TFTN model and the input is the pseudo-color image corresponding to hyperspectral data (TFTN_single_HSI). Besides, we also compare the performance of the model when the input of one branch structure of the TFTN model is RGB data, and the input of the other branch is hyperspectral data (TFTN_dual_RGB-HSI).

Specifically, each branch of the TFTN_dual_RGB model is trained by RGB data. Due to the channels of RGB data and hyperspectral data are different, we have improved the parameters of the HSI backbone originally designed for hyperspectral modality data, which makes the network suitable for RGB data while ensuring consistency with the original parameters as much as possible, as shown in Table V. Compared with the TFTN_dual_RGB model, the TFTN_single_RGB model only contains the RGB branch in the TFTN model, which is equivalent to the single-modality Siamese tracker. In addition, the input of the RGB branch of the TFTN_dual_HSI model is the pseudo-color image corresponding to hyperspectral data, and the input of the HSI branch is hyperspectral data. Besides, the TFTN_dual_RGB-HSI model is the proposed

TABLE V

PARAMETERS OF CONVOLUTIONAL LAYERS IN THE HSI BACKBONE OF THE TFTN DUAL RGB MODEL

Block Name	Kernel Name	Kernel Number	Kernel Size (H,W,B)	Kernel Stride Δ(H,W,B)
Block #1	conv1	256	7,7,3	2,2,1
	conv1_1	256	3,3,1	2,2,1
	conv1_2	256	3,3,1	2,2,1
Block #2	conv2_1	256	1,1,1	1,1,1
	conv2_2	256	1,1,3	
Block #3	conv3_1	256	1,1,1	1,1,1
	conv3_2	256	1,1,3	

The symbol B denotes the kernel size in the spectral dimension.

TABLE VI

RESULTS OF THE TFTN SINGLE RGB TRACKER, TFTN DUAL RGB TRACKER, TFTN SINGLE HSI TRACKER, TFTN DUAL HSI TRACKER, AND TFTN DUAL RGB-HSI TRACKER IN TERMS OF AUC SCORE AND DP 20 VALUE

	AUC	Δ(AUC)	DP_20	Δ(DP_20)
TFTN_single_RGB	61.3%	-	84.1%	-
TFTN_dual_RGB	72.7%	↑ 11.4%	93.0%	↑ 8.9%
TFTN_dual_RGB-HSI	76.5%	↑ 15.2%	96.2%	↑ 12.1%
TFTN_single_HSI	60.5%	-	87.4%	-
TFTN_dual_HSI	64.6%	↑ 4.1%	87.5%	↑ 0.1%
TFTN_dual_RGB-HSI	76.5%	↑ 16.0%	96.2%	↑ 8.8%

complete TFTN model, in which two branches are trained by hyperspectral and RGB data pairs.

The results of the TFTN_dual_RGB tracker, the TFTN_single_RGB tracker, and the TFTN_dual_RGB-HSI tracker are compared in Table VI. It can be seen that the AUC score of the TFTN_dual_RGB tracker (72.7%) outperforms the AUC score of the TFTN_single_RGB tracker (61.3%) by 11.4%, and the DP_20 value of the TFTN_dual_RGB tracker (93.0%) is more than the DP_20 value of the TFTN_single_RGB tracker (84.1%) by 8.9%. The experimental results show that the fusion tracking model with the dual-branch structure shows better performance than the tracking model with the single-branch structure, which fully demonstrates the effectiveness of the proposed fusion framework. In addition, it can also be seen that the AUC score increases from 72.7% to 76.5% after using hyperspectral data, and the DP_20 value is improved from 93.0% to 96.2%, which shows that the performance of the fusion tracking task can be further improved by using hyperspectral data information. Table VI also shows that the AUC score of the TFTN_dual_HSI tracker (64.6%) outperforms that of the TFTN_single_HSI tracker (60.5%) by 4.1%, and the DP_20 value of the TFTN_dual_HSI tracker (87.5%) is higher than that of the TFTN_single_HSI tracker (87.4%) by 0.1%, again verifying the effectiveness of the proposed fusion framework. Besides, after using RGB data to replace pseudo-color data, the AUC score of the TFTN_dual_HSI model increases from 64.5% to 76.5%, and the DP_20 value increases from 87.5%

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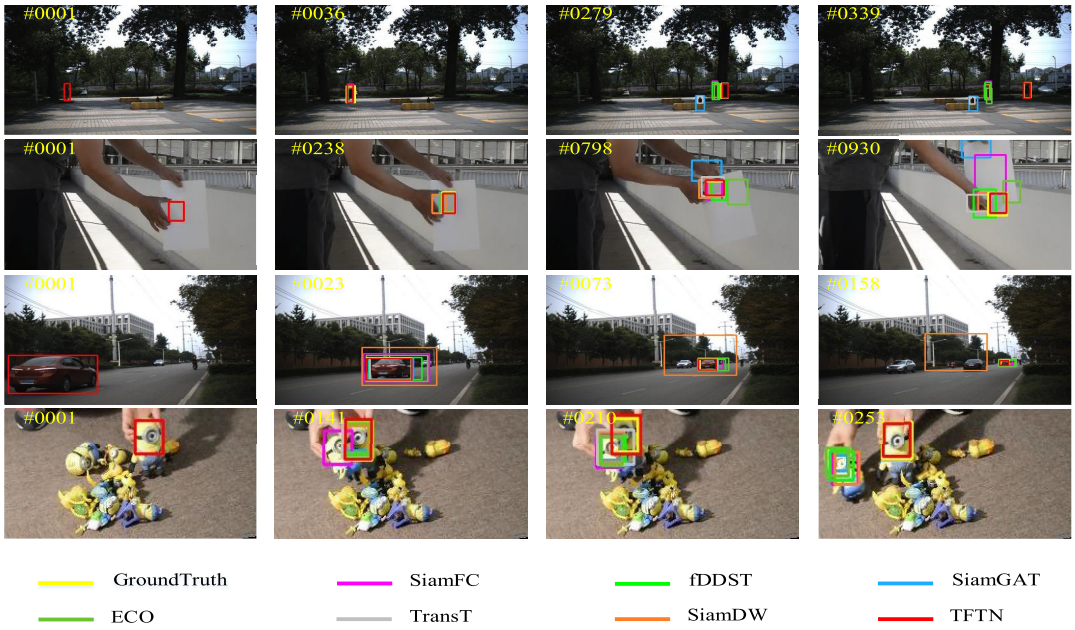


Fig. 10. Qualitative comparison of seven trackers and groundtruth on some sequences in some challenging scenarios. The sequence names from top to bottom are Pedestrian2, Card, Car3, and Toy1.

TABLE VII
RESULTS OF THE CONCAT_FUSION TRACKER, CROSS_FUSION TRACKER, AND TFTN TRACKER IN TERMS OF AUC SCORE, DP_20 VALUE AND FRAMES PER SECOND (FPS)

	Concat_fusion	Cross_fusion	TFTN
AUC	63.7%	63.3%	76.5%
DP_20	86.8%	86.2%	96.2%
FPS	14.1	14.3	17.2

to 96.2%, indicating that the visual information provided by RGB data helps to improve the performance of fusion tracking tasks.

In addition, we also supplement the comparative experiments with other fusion frameworks. We train and test the performance of two typical fusion frameworks and compare the performance of our method with that of them. Among them, one fusion framework adopted by Zhu et al. [51] uses the concatenation-based fusion method to combine multimodality information, termed Concat_fusion. And the other fusion framework proposed by Zhang et al. [52] seeks to learn more compact feature representations across modalities by interactively concatenating the depth features from different modalities, termed Cross_fusion. The results are shown in Table VII. It can be seen that the AUC score of the TFTN tracker (76.5%) outperforms that of the Concat_fusion tracker (63.7%) and Cross_fusion tracker (63.3%), and the DP_20 value of the TFTN tracker (96.2%) is more than that of the Concat_fusion tracker (86.8%) and Cross_fusion tracker (86.2%). The above results show that the proposed TFTN tracker outperforms the tracker based on other fusion frameworks, confirming the superiority and effectiveness of the proposed Transformer-based fusion framework. In addition, it can also be seen from the table that the proposed tracker based on the Transformer fusion strategy has the fastest speed compared with that of the tracking algorithm based on other fusion strategies, which confirms the effectiveness of the tracker based on the Transformer fusion strategy again.

C. Experiment3: Effectiveness of the Siamese 3-D CNN Branch

The designed Siamese 3-D CNN branch consists of two multiscale 3-D CNNs and a depth-wise cross correlation layer, which are used to obtain hyperspectral MS representations. We design four groups of subexperiments to verify the effectiveness of different components in the Siamese 3-D CNN, respectively.

First, the importance of spatial–spectral joint information is evaluated to verify the effectiveness of the 3-D CNN in this branch. Compared with the RGB image, the HSI has a 3-D structure that includes 2-D space and rich 1-D spectral information. To fully capture the characteristics of hyperspectral data, we use 3-D convolution kernels to slide together in spatial and spectral dimensions of hyperspectral data, which can obtain more complex spatial–spectral joint information. The obtained spatial–spectral joint information is the MS information of hyperspectral data, which can provide more powerful supplements for RGB visual information. To evaluate the importance of spatial–spectral joint information for improving the performance of the fusion tracking algorithm based on hyperspectral and RGB data, we compare the performance of the TFTN model which uses 3-D CNN to obtain the spatial–spectral joint information of hyperspectral modalities (TFTN_3-D) with that of the TFTN model which uses 2-D CNN instead of 3-D CNN to obtain the spatial information of

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TABLE VIII

PARAMETERS OF CONVOLUTIONAL LAYERS IN THE HSI BACKBONE OF THE TFTN_2-D MODEL

Block Name	Kernel Name	Kernel Number	Kernel Size (H,W)	Kernel Stride Δ(H,W)
Block #1	conv1	256	7,7	2,2
	conv1_1	256	3,3	2,2
	conv1_2	256	3,3	2,2
Block #2	conv2_1	256	1,1	1,1
	conv2_2	256	1,1	
Block #3	conv3_1	256	1,1	1,1
	conv3_2	256	1,1	

TABLE IX

RESULTS OF THE TFTN_2-D TRACKER AND TFTN_3-D TRACKER IN TERMS OF AUC SCORE AND DP_20 VALUE

	AUC	Δ(AUC)	DP_20	Δ(DP_20)
TFTN_2D	73.1%	-	94.8%	-
TFTN_3D	76.5%	↑ 3.4%	96.2%	↑ 1.4%

hyperspectral data (TFTN_2-D). Among them, the TFTN_3-D model is the proposed complete TFTN model. Besides, the parameters in the HSI backbone of the TFTN_2-D model are shown in Table VIII.

The results of the TFTN_3-D tracker and the TFTN_2-D tracker are compared in Table IX. It can be seen that the AUC score of the TFTN_3-D tracker (76.5%) outperforms the AUC score of the TFTN_2-D tracker (73.1%) by 3.4% after using 3-D CNN, and the DP_20 value of the TFTN_3-D tracker (96.2%) is more than the DP_20 value of the TFTN_2-D tracker (94.8%) by 1.4%. The results show that the spatial–spectral joint information of hyperspectral data is more beneficial to improve the performance of the fusion tracking algorithm based on hyperspectral and RGB data than the spatial information of hyperspectral data, indicating that the 3-D CNN in the branch is effective.

Then, by evaluating the importance of multiscale information to verify the effectiveness of the multiscale structure in the branch. On the basis of using 3-D CNN to extract spatial–spectral joint information of hyperspectral modalities, we propose a 3-D CNN with a multiscale structure, which enables the tracking network to capture spectral multiscale objects in the spatial domain of HSIs adaptively. To verify the importance of the obtained multiscale spectral information, we also design two TFTN models with a single-scale structure that only can capture single-scale spectral information (one is termed as TFTN_single1 and the other is termed as TFTN_single2) and compare their performance with that of the TFTN model with the multiscale structure (TFTN_multiscale). Among them, the TFTN_multiscale model is the proposed complete TFTN model. The HSI backbone structure of the TFTN_single1 model is shown in Fig. 11, and that of the TFTN_single2 model is shown in Fig. 12.

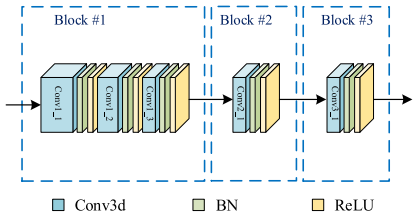


Fig. 11. HSI backbone structure of the TFTN_single1 model.

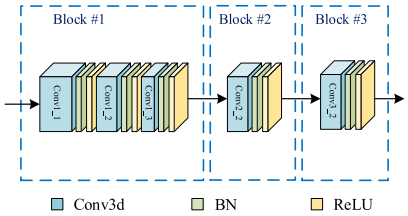


Fig. 12. HSI backbone structure of the TFTN_single2 model.

TABLE X

RESULTS OF THE TFTN_MULTISCALE TRACKER, TFTN_SINGLE1 TRACKER, AND TFTN_SINGLE2 TRACKER IN TERMS OF AUC SCORE AND DP_20 VALUE

	AUC	Δ(AUC)	DP_20	Δ(DP_20)
TFTN_single1	74.4%	-	93.7%	-
TFTN_multi-scale	76.5%	↑ 2.1%	96.2%	↑ 2.5%
TFTN_single2	74.5%	-	93.8%	-
TFTN_multi-scale	76.5%	↑ 2.0%	96.2%	↑ 2.4%

In Table X, it can be seen that the AUC score of the TFTN_multiscale tracker (76.5%) outperforms that of the TFTN_single1 tracker (74.4%) by 2.1%, and the DP_20 value of the TFTN_multiscale tracker (96.2%) is more than that of the TFTN_single1 tracker (93.7%) by 2.5%. It can also be seen that the TFTN_multiscale tracker outperforms the TFTN_single2 tracker in terms of AUC score and DP_20 value. The above results show that the multiscale spectral information of the hyperspectral modality is conducive to improving the performance of the fusion tracking task based on hyperspectral and RGB data, indicating that the multiscale structure in the Siamese 3-D CNN is effective.

Third, evaluate the importance of transmitting multilevel semantic information to verify the effectiveness of the depth-wise cross correlation layer in the branch. Since different levels of features have different semantic information, to more fully transmit the template patch information of hyperspectral modality to the search region, we need to consider the information interaction of different level features between the template patch and the search region. Therefore, we use the depth-wise cross correlation layer composed of two single-level cross correlation operations to calculate the correlation of the features extracted from Block #2 and Block #3 of the HSI backbone to obtain richer HSI MS information. To verify the importance of transmitting multilevel semantic information, we compare the performance of the TFTN model

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TABLE XI
RESULTS OF THE TFTN_SCCL TRACKER AND TFTN_DCCL TRACKER IN TERMS OF AUC SCORE AND DP_20 VALUE

	AUC	Δ(AUC)	DP_20	Δ(DP_20)
TFTN_sccl	73.4%	-	93.0%	-
TFTN_dccl	76.5%	↑ 3.1%	96.2%	↑ 3.2%

TFTN_dccl is the TFTN model with the depth-wise cross-correlation layer.

TABLE XII
ABLATION STUDY OF THE BLOCK IN THE SIAMESE 3-D CNN

Block Name	AUC	Δ(AUC)	DP_20	Δ(DP_20)
B#1	72.6%	-	92.1%	-
B#1 + B#2	74.0%	↑ 1.4%	93.1%	↑ 1.0%
B#1 + B#2 + B#3	76.5%	↑ 3.9%	96.2%	↑ 4.1%

B#1, B#2, and B#3 represent Block #1, Block #2, and Block #3, respectively.

(TFTN_sccl) that only uses a single cross correlation layer to transmit HSI single-level semantic information extracted from HSI backbone Block #3 with that of the TFTN model (TFTN_dccl) that uses the deep-wise cross correlation layer to transmit HSI multilevel semantic information. Among them, the TFTN_dccl model is the proposed complete TFTN model.

It can be seen from Table XI that the AUC score increases from 73.4% to 76.5% after using the depth-wise cross correlation layer, and the DP_20 value is improved from 93.0% to 96.2%. The results show that using the deep-wise cross correlation layer to transmit multilevel semantic information in the Siamese 3-D CNN can effectively improve the performance of the fusion tracking task, indicating that the deep-wise cross correlation layer in the Siamese 3-D CNN is effective.

Finally, we also verify the effect of each block of the Siamese 3-D CNN on the final performance. The ablation study results are listed in Table XII. The model in which Siamese 3-D CNN contains Block #1, Block #2, and Block #3 in Table XII is the complete TFTN model. It can be seen that using three blocks in the Siamese 3-D CNN is 3.9% higher than using only Block #1 in terms of the AUC score and 4.1% higher in terms of the DP_20 value.

The results show that the proposed Siamese 3-D CNN with three blocks can obtain the semantic information of the hyperspectral modality with more resolution, which can effectively improve the performance of the fusion tracking task.

D. Experiment 4: Generic of the Fusion Framework

The proposed fusion tracking framework mainly includes two branches, which can be used to process RGB and hyperspectral data, respectively. Each branch is designed based on the Siamese network, which can be flexibly replaced by other Siamese tracking structures. Because of the fast iteration of the Siamese tracking algorithms with RGB video data as the research object, the proposed fusion tracking algorithm can achieve better tracking performance if the specific branch of the RGB modality has a better performance. To validate the generality of the fusion framework, we train and test the performance of the TFTN model which uses other RGB Siamese tracking structures to replace the RGB MS branch.

TABLE XIII
RESULTS OF THE TFTN_GAT TRACKER AND SAIMGAT TRACKER IN TERMS OF AUC SCORE AND DP_20 VALUE

	AUC	Δ(AUC)	DP_20	Δ(DP_20)
SiamGAT	63.6%	-	86.4%	-
TFTN_GAT	66.9%	↑ 3.3%	91.6%	↑ 5.2%

In Table XIII, TFTN_graph attention tracking (GAT) indicates that the RGB MS branch is replaced by the TFTN model designed according to the SiamGAT algorithm. The RGB MS branch of TFTN_GAT uses the modified GoogLeNet [53] as the backbone to extract the features of RGB data and disseminates the modality information between the template patch and search region through the graph attention mechanism.

It can be seen that the AUC score of the TFTN_GAT tracker (66.9%) outperforms that of the SiamGAT tracker (63.6%), and the DP_20 value of the TFTN_GAT tracker (91.6%) is more than that of the SiamGAT tracker (86.4%). The experimental results show that the proposed fusion framework is generic, and the replaceable branches provide infinite possibilities for future use of this framework to improve the performance of the fusion tracking task.

VI. CONCLUSION

In this article, we propose a novel fusion tracking framework of hyperspectral and RGB based on Transformer, termed as TFTN, aiming to improve the tracking performance by adding spectral information to the RGB modality. The framework is generic, which is suitable for the Siamese series of tracking algorithms. A dual-branch structure based on the Siamese network is used to obtain the MS representations of the two modalities, in which the Siamese 3-D CNN is designed as the specific branch of hyperspectral modality to synchronously extract the spatial and spectral features of hyperspectral data. Then the designed TFM is used to adaptively capture the potential interaction within the modalities and between the modalities of the multimodality images, forming effective modality fusion representations. The fusion result is used to enhance the classification and regression ability of the tracking network. The comparison of experimental results with state-of-the-art trackers on challenging sequences demonstrates the effectiveness of the proposed method.

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REFERENCES

[1] X. Tian, J. Liu, M. Mallick, and K. Huang, "Simultaneous detection and tracking of moving-target shadows in ViSAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1182–1199, Feb. 2021.

[2] X. Yang, Y. Wang, N. Wang, and X. Gao, "An enhanced SiamMask network for coastal ship tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[3] M. Thomas, C. Kambhamettu, and C. A. Geiger, "Motion tracking of discontinuous sea ice," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5064–5079, Dec. 2011.

[4] X. Zhang, X. Zhang, X. Du, X. Zhou, and J. Yin, "Learning multi-domain convolutional network for RGB-T visual tracking," in *Proc. 11th Int. Congr. Image Signal Process., Biomed. Eng. Informat. (CISP-BMEI)*, Oct. 2018, pp. 1–6.

[5] X. Zhang, P. Ye, S. Peng, J. Liu, K. Gong, and G. Xiao, "SiamFT: An RGB-infrared fusion tracking method via fully convolutional Siamese networks," *IEEE Access*, vol. 7, pp. 122122–122133, 2019.

[6] X. Zhang, P. Ye, S. Peng, J. Liu, and G. Xiao, "DSiamMFT: An RGB-T fusion tracking method via dynamic Siamese networks using multi-layer feature fusion," *Signal Process., Image Commun.*, vol. 84, May 2020, Art. no. 115756.

[7] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Conf. North Amer. Chapter Assoc. Comput. Linguistics: Hum. Lang. Technol. (NAACL-HLT)*, vol. 1, 2019, pp. 4171–4186.

[9] B. Satar, Z. Hongyuan, X. Bresson, and J. H. Lim, "Semantic role aware correlation transformer for text to video retrieval," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 1334–1338.

[10] Y. Zhu, S. Wang, Z. Huang, and K. Chen, "Text recognition in images based on transformer with hierarchical attention," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1945–1949.

[11] T. Le, H. T. Nguyen, and M. L. Nguyen, "Vision and text transformer for predicting answerability on visual question answering," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 934–938.

[12] D. Henke, E. M. Dominguez, D. Small, M. E. Schaepman, and E. Meier, "Moving target tracking in single- and multichannel SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3146–3159, Jun. 2015.

[13] T. Jiang, Z. Zhang, Y. Xu, Y. Bai, and Y. Wang, "First-person multiple object tracking in complex traffic scenes," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2014, pp. 2343–2347.

[14] Y. Han, C. Deng, Z. Zhang, J. Li, and B. Zhao, "Adaptive feature representation for visual tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 1867–1870.

[15] Q. Guo, W. Feng, R. Gao, Y. Liu, and S. Wang, "Exploring the effects of blur and deblurring to visual object tracking," *IEEE Trans. Image Process.*, vol. 30, pp. 1812–1824, 2021.

[16] F. Gurkan, L. Cerkezi, O. Cirakman, and B. Gunsel, "TDIOT: Target-driven inference for deep video object tracking," *IEEE Trans. Image Process.*, vol. 30, pp. 7938–7951, 2021.

[17] S. Xuan, S. Li, M. Han, X. Wan, and G. S. Xia, "Object tracking in satellite videos by improved correlation filters with motion estimations," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1074–1086, Feb. 2020.

[18] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.

[19] J. Shao, B. Du, C. Wu, and L. Zhang, "Can we track targets from space? A hybrid kernel correlation filter tracker for satellite video," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8719–8731, Aug. 2019.

[20] M. M. Dasari and R. K. S. S. Gorthi, "IOU—Siamtrack: IOU guided siamese network for visual object tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2061–2065.

[21] C. Fu, Z. Cao, Y. Li, J. Ye, and C. Feng, "Onboard real-time aerial tracking with efficient Siamese anchor proposal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[22] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. Torr, "Fully-convolutional siamese networks for object tracking," in *Proc. Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science). Amsterdam, The Netherlands, 2016, pp. 850–865.

[23] J. Shao, B. Du, C. Wu, M. Gong, and T. Liu, "HRSiam: High-resolution Siamese network, towards space-borne satellite video tracking," *IEEE Trans. Image Process.*, vol. 30, pp. 3056–3068, 2021.

[24] M. H. Abdelpakey and M. S. Shehata, "DP-siam: Dynamic policy Siamese network for robust object tracking," *IEEE Trans. Image Process.*, vol. 29, pp. 1479–1492, 2020.

[25] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with Siamese region proposal network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8971–8980.

[26] Y. Xu, Z. Wang, Z. Li, Y. Yuan, and G. Yu, "SiamFC++: Towards robust and accurate visual tracking with target estimation guidelines," in *Proc. Conf. Artif. Intell. (AAAI)*, 2020, vol. 34, no. 7, pp. 12549–12556.

[27] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "SiamRPN++: Evolution of Siamese visual tracking with very deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4277–4286.

[28] D. Guo, Y. Shao, Y. Cui, Z. Wang, L. Zhang, and C. Shen, "Graph attention tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9543–9552.

[29] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, and H. Lu, "Transformer tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8126–8135.

[30] D. Guo, J. Wang, Y. Cui, Z. Wang, and S. Chen, "SiamCAR: Siamese fully convolutional classification and regression for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6268–6276.

[31] Z. Zhang and H. Peng, "Deeper and wider Siamese networks for real-time visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4586–4595.

[32] G. Tochon, J. Chanussot, M. D. Mura, and A. L. Bertozzi, "Object tracking by hierarchical decomposition of hyperspectral video sequences: Application to chemical gas plume tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4567–4585, May 2017.

[33] K. Qian, J. Zhou, F. Xiong, H. Zhou, and J. Du, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," 2018, *arXiv:1810.11819*.

[34] H. Van Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (Workshops)*, Jun. 2010, pp. 44–51.

[35] B. Uzkent, M. J. Hoffman, and A. Vodacek, "Real-time vehicle tracking in aerial video using hyperspectral features," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 1443–1451.

[36] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[37] Z. Liu et al., "An anchor-free Siamese target tracking network for hyperspectral video," in *Proc. 11th Workshop Hyperspectral Imaging Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[38] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "BAE-Net: A band attention aware ensemble network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2106–2110.

[39] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[40] X. Lan, W. Zhang, S. Zhang, D. K. Jain, and H. Zhou, "Robust multi-modality anchor graph-based label prediction for RGB-infrared tracking," *IEEE Trans. Ind. Informat.*, early access, Oct. 14, 2019, doi: 10.1109/TII.2019.2947293.

[41] X. Lan, M. Ye, R. Shao, B. Zhong, P. C. Yuen, and H. Zhou, "Learning modality-consistency feature templates: A robust RGB-infrared tracking system," *IEEE Trans. Ind. Electron.*, vol. 66, no. 12, pp. 9887–9897, Dec. 2019.

[42] X. Zhang, P. Ye, D. Qiao, J. Zhao, S. Peng, and G. Xiao, "Object fusion tracking based on visible and infrared images using fully convolutional Siamese networks," in *Proc. 22th Int. Conf. Inf. Fusion (FUSION)*, Jul. 2019, pp. 1–8.

[43] N. Xu, G. Xiao, X. Zhang, and D. P. Bavirisetti, "Relative object tracking algorithm based on convolutional neural network for visible and infrared video sequences," in *Proc. 4th Int. Conf. Virtual Reality (ICVR)*, 2018, pp. 44–49.

[44] J. Lanchantin, T. Wang, V. Ordonez, and Y. Qi, "General multi-label image classification with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16473–16483.

[45] M. Li, J. Liu, C. Zheng, X. Huang, and Z. Zhang, "Exploiting multi-view part-wise correlation via an efficient transformer for vehicle re-identification," *IEEE Trans. Multimedia*, early access, Dec. 13, 2021, doi: 10.1109/TMM.2021.3134839.

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[46] C. Zou et al., "End-to-end human object interaction detection with hoi transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11820–11829.
[47] V. Gabeur, C. Sun, A. Karteek, and C. Schmid, "Multi-modal transformer for video retrieval," in *Proc. Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science). Glasgow, U.K., 2020.
[48] L. Liu, W. Wang, Z. Zhang, M. Zhangz, P. Pengz, and X. Sunz, "Integrated modalities and multi-level granularity: Towards a unified video-text retrieval framework," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2021, pp. 1–2.
[49] Z. Liu, Y. Tan, Q. He, and Y. Xiao, "SwinNet: Swin transformer drives edge-aware RGB-D and RGB-T salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 7, pp. 4486–4497, Jul. 2022.
[50] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, pp. 211–252, Apr. 2015.
[51] Y. Zhu, C. Li, B. Luo, J. Tang, and X. Wang, "Dense feature aggregation and pruning for RGBT tracking," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 465–472.
[52] L. Zhang, M. Danelljan, A. Gonzalez-Garcia, J. van de Weijer, and F. Shahbaz Khan, "Multi-modal fusion for end-to-end RGB-T tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2252–2261.
[53] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.



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